Bézout Domain and Noetherian Domain

Chong Jongwon

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University of Seoul, Department of Mathematics

Recall: Definitions in Ring

Definition
A Commutative Ring R with 1 is called an **Integral Domain** if:

$$R \text{ has no zero divisor.}$$

That is, for any $a, b \in R$, $ab = 0$ implies either $a = 0$ or $b = 0$.

Definition
In an integral domain R, a subring $I \subset R$ is called an **Ideal** if:

$$\text{Given } r \in R \text{ and } a \in I,\ ra \text{ is contained in } I$$

Proposition
Let R be an integral domain, and $A \subset R$ be a subset. Then,

$$(A) \overset{\text{def}}{=} \bigcap_{\substack{A \subset I \\ I \text{ is ideal}}} I = RA = \{r_1 a_1 + \cdots + r_n a_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$

Bézout Domain

Definition (Bézout Domain)
An integral domain is called a **Bézout Domain** if it satisfies:

$$\text{for any } a, b \in R, \text{ there exists } d \in R \text{ such that } (a, b) = (d)$$

Example
In the polynomial ring $\mathbb{Z}[x]$, 2 and x both are divided by 1, but

$$(2, x) \neq (d)$$

for all $d \in \mathbb{Z}[x]$. Thus, $\mathbb{Z}[x]$ is **Not** a Bézout domain.

Bézout Domain: Every finitely generated ideal is principal

Theorem
Let R be an integral domain. Then,

R is a Bézout Domain
if and only if
Every finitely generated ideal in R is principal

Noetherian Domain

Noetherian Domain

Definition (Noetherian Domain)
An integral domain R is called a **Noetherian Domain** if it satisfies:
For every sequence of ideals $\{I_n\}_{n=1}^{\infty}$ such that

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

, there exists $N \in \mathbb{N}$ such that $I_n = I_N$ for all $n \geq N$.

Such a condition is called the **Ascending Chain Condition** on ideals.

Noetherian Domain: Every ideal is finitely generated

Theorem
Let R be an integral domain. Then,

R is a Noetherian Domain

if and only if

Every ideal in R is finitely generated.

Principal Ideal Domain

Principal Ideal Domain

Definition
An integral domain R is called a **Principal Ideal Domain** if:

Every ideal in R is principal

Theorem
Let R be an integral domain. Then:

R **is a Principal Ideal Domain**
if and only if
R **is Bézout Domain and Noetherian Domain**

End